//TCP Server

```
#include<stdlib.h>
#include<stdio.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main()
{
        char buff[512];
        int SocketFD;
        struct sockaddr_in stSockAddr, cli_addr;
        SocketFD=socket(AF_INET,SOCK_STREAM,0);
if(SocketFD<0)
        {
                perror("can not creat socket");
exit(EXIT_FAILURE);
        }
        bzero(&stSockAddr,sizeof(stSockAddr));
stSockAddr.sin_family=AF_INET;
stSockAddr.sin_port=htons(1109);
stSockAddr.sin_addr.s_addr=INADDR_ANY;
        if(bind(SocketFD,(struct sockaddr*)&stSockAddr,sizeof(stSockAddr))<0)
        {
                perror("can not bind");
close(SocketFD);
                exit(EXIT_FAILURE);
        }
        int a=sizeof(cli_addr);
        listen(SocketFD,2);

        int connectFD=accept(SocketFD,(struct sockaddr*)&cli_addr,&a);
        if(connectFD<0)
        {
                perror("can not connect");
close(SocketFD);
                exit(EXIT_FAILURE);
        }

        while(1)
        {
                printf("\n Receiving data from client.....");
        read(connectFD,buff,512);
```

```
                if(strcmp(buff,"x")==0)
                        break;
                printf("\n data is:%s",buff);

                printf("\nenter data to be
sent");                 //scanf("%s",buff);
gets(buff);
                write(connectFD,buff,512);
        }
        close(connectFD);
printf("\n");
        return EXIT_SUCCESS;
}
```

//TCP Client

```
#include<stdlib.h>
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main()
{
        char buff[512];
int SocketFD;
        struct sockaddr_in stSockAddr;
        SocketFD=socket(AF_INET,SOCK_STREAM,0);
if(SocketFD<0)
        {
                perror("can not creat socket");
exit(EXIT_FAILURE);
        }
        bzero(&stSockAddr,sizeof(stSockAddr));
stSockAddr.sin_family=AF_INET;
stSockAddr.sin_port=htons(1109);
        stSockAddr.sin_addr.s_addr=inet_addr("127.0.0.1");
        int xyz=connect(SocketFD,(struct
sockaddr*)&stSockAddr,sizeof(stSockAddr));        if(xyz<0)
        {
```

```
                perror("can not connect");
close(SocketFD);
                exit(EXIT_FAILURE);
        }
        while(1)
        {
                printf("\nenter data to be
sent");                 //scanf("%s",buff);
gets(buff);
write(SocketFD,buff,512);

                printf("\n Receiving data from Server.....");
        read(SocketFD,buff,512);

                if(strcmp(buff,"x")==0)
                        break;
                printf("\n data is:%s",buff);
        }
        printf("\n");
        close(SocketFD);
}
```

//TCP Server- Reverse String

```
#include<stdlib.h>
#include<stdio.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main() {  char buff[512],str1[512],c;  int
SocketFD, len, i,k;  struct sockaddr_in
stSockAddr, cli_addr;
 SocketFD=socket(AF_INET,SOCK_STREAM,0);
 if(SocketFD<0)
 {      perror("can not creat socket");
exit(EXIT_FAILURE);
 }  bzero(&stSockAddr,sizeof(stSockAddr));
stSockAddr.sin_family=AF_INET;
stSockAddr.sin_port=htons(1504);
stSockAddr.sin_addr.s_addr=INADDR_ANY;
 if(bind(SocketFD,(struct sockaddr*)&stSockAddr,sizeof(stSockAddr))<0)
```

```
 {        perror("can not bind");
close(SocketFD);
            exit(EXIT_FAILURE);
 }
 int a=sizeof(cli_addr);  listen(SocketFD,2);

 int connectFD=accept(SocketFD,(struct sockaddr*)&cli_addr,&a);  if(connectFD<0)
 {
          perror("can not connect");      close(SocketFD);
            exit(EXIT_FAILURE);
 }
          printf("\n Received String:");
read(connectFD,buff,512);
            printf("%s",buff);
             len=strlen(buff);

            for(i=len-1, k=0; i>=0; i--, k++)
            {
                      c=buff[i];
                    str1[k]=c;
}
            //strrev(buff);
            printf("\nReversed String:%s", str1);

            write(connectFD,str1,512);
   close(connectFD);
printf("\n");
 return EXIT_SUCCESS;
}
```

//TCP Client- Reverse String

```
#include<stdlib.h>
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main() {  char
buff[512];  int
SocketFD;
 struct sockaddr_in stSockAddr;
```

```
 SocketFD=socket(AF_INET,SOCK_STREAM,0);
 if(SocketFD<0)
 {        perror("can not creat socket");
exit(EXIT_FAILURE);
 }
 bzero(&stSockAddr,sizeof(stSockAddr));
stSockAddr.sin_family=AF_INET;  stSockAddr.sin_port=htons(1504);
 stSockAddr.sin_addr.s_addr=inet_addr("127.0.0.1");
 int xyz=connect(SocketFD,(struct sockaddr*)&stSockAddr,sizeof(stSockAddr));
if(xyz<0)  {     perror("can not connect");      close(SocketFD);
            exit(EXIT_FAILURE);
 }

            printf("\nEnter String to be sent");
           gets(buff);       write(SocketFD,buff,512);
printf("\n Reversed String=");
            read(SocketFD,buff,512);

            printf("%s",buff);
   printf("\n");
close(SocketFD);
}
```